Course- BTech.
Course Code- CSET343
Year- 4TH Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment - 9

Objective:

To design and evaluate a 3D Convolutional Neural Network (3D CNN) for automated fall detection using video data. The experiment focuses on identifying falls in controlled environments to support elderly care and hospital monitoring systems.

Dataset:

https://www.kaggle.com/datasets/shahliza27/ur-fall-detection-dataset

Description (URFD): (4 GB)
• Total Videos: 70 sequences (30 falls, 40 ADL)
• Modalities: RGB + Depth (Kinect) with accelerometer streams
• Typical Resolution: 320×240 pixels
• Application: Fall detection in healthcare and assisted living

Tasks:

Task 1: Data Acquisition and Exploration

- Download the dataset from any working mirror and explore video samples.
- Differentiate between normal activity and fall sequences visually.

Task 2: Data Preprocessing

- Extract a fixed number of frames from each video (e.g., 16 or 32).
- Resize each frame to a fixed dimension (e.g., 112 × 112 pixels).
- Normalize pixel intensity values to the range [0, 1].
- Convert to RGB if required (ensure consistent color channels).
- Stack frames sequentially to form 3D video tensors (shape: *frames* × *height* × *width* × *channels*).
- Apply data augmentation techniques:
  - Random horizontal flipping
  - Random cropping or rotation
  - Temporal frame jittering (vary start frame)
- Handle class imbalance by oversampling minority classes or applying weighted loss.
- Verify and remove corrupted video files before training.
- Split data into **training (70%)**, **validation (15%)**, and **testing (15%)**.

Task 3: Model Training and Basic Evaluation

- Implement **baseline models**: a 3D CNN for classifying fall vs. non-fall activities
- Evaluate models using **Accuracy, Precision, Recall, F1-score, and AUROC**.
- Analyze confusion matrix and ROC curves; identify common misclassifications.
- Relate errors to camera view, occlusion, or motion blur.